Endpoint Composition Shift: A New Class of Distribution Shift, with PAC-Bayes Generalisation Bounds and a Regime-Aware Self-Routing Network for Longitudinal Medical AI

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Abstract

Medical-AI leaderboards routinely treat model rankings as algorithmic properties. We formalise a previously-uncharacterised class of distribution shift — **Endpoint Composition Shift (ECS)** — in which the source and target cohorts share per-stratum conditional distributions but differ in the prior over biologically-meaningful strata. ECS is provably distinct from generic label shift: the stratification is *pre-deployment estimable* from clinical protocol metadata, and existing label-shift estimators (BBSE, RLLS, MLLS, ATC, PAPE, Saerens-EM) do not exploit this structure. We develop a complete theoretical framework: **(Theorem 2)** ECS as a formal class of shift; **(Theorem 3)** PAC-Bayes generalisation bound on ranking-reversal probability with explicit dependence on per-stratum sample size; **(Theorem 4)** Bayes-optimal regret bound for π\-*thresholded routing; **(Theorem 5)** conformal coverage guarantee for three-regime classification. We then introduce **RASN — a Regime-Aware Self-Routing Network** — a differentiable ensemble that estimates the cohort π_stable from input batches and routes each case between a heat-kernel prior path and a learned 3D U-Net path via a calibrated soft-router. Across 4 genuinely independent cohorts (UCSF, MU-Glioma-Post, RHUH-GBM, UCSD-PTGBM; N=522), 3 seeds × 4 cohorts × 12 train/test conditions, RASN delivers regret competitive with the per-cohort Bayes-optimal model selector. Five seeds × two architectures × four cohorts × five model variants from prior work preserve all directional outcomes (20/20). The closed-form crossover threshold* π\=0.43 (95% CI [0.30, 0.52]) and its Bayesian credible interval [0.17, 0.59] are robust to per-stratum re-estimation. Yale label-free acquisition-shift screening (AUROC=0.847) provides a complementary deployment audit. RASN realises a "regime-as-architecture" design pattern in which a known deployment-context structure becomes a differentiable architectural constraint — a pattern that extends in principle to any distributional-shift problem with known stratification.

Keywords: endpoint composition shift; PAC-Bayes generalisation; longitudinal medical AI; regime-aware ensemble; conformal prediction; benchmark transportability; closed-form crossover; calibrated routing.

1. Introduction

1.1 The benchmark transportability problem in longitudinal medical AI

Medical-AI leaderboards rank models on fixed cohorts and treat the resulting ordering as if it were a stable property of the algorithms compared. In longitudinal neuro-oncology, this assumption fails systematically. A model evaluated on a surveillance-dominant post-operative GBM cohort (where most paired evaluations show stable disease) ranks differently from the same model evaluated on an active-change-enriched radiotherapy-planning cohort. The principal mechanism is not algorithmic: it is the *endpoint composition* of the cohort.

Roberts et al. (2021) demonstrated that none of 62 published COVID models was clinically usable, with cohort selection driving apparent winners. Maier-Hein et al. (Metrics Reloaded, 2024) identified rank sensitivity as the dominant failure mode across 150 challenges. Karargyris et al. (MedPerf, 2023) built federated infrastructure to *measure* per-site heterogeneity, but the field has lacked a formal theoretical framework for predicting ranking instability *before deployment*.

1.2 Existing label-shift theory does not address the problem

Label-shift methods (Saerens et al. 2002; BBSE, Lipton et al. 2018; RLLS, Azizzadenesheli et al. 2019; MLLS, Alexandari et al. 2020; ATC, Garg et al. 2022; PAPE, Garg et al. 2025) all require *target-domain unlabeled data* to estimate the corrected loss, and they address single-model correction rather than inter-model crossover. None of these methods provides a closed-form, pre-deployment ranking-reversal predictor that uses only source-cohort statistics.

Within longitudinal medical AI, no existing methodology formalises the problem in a way that admits PAC-Bayes generalisation bounds, conformal coverage guarantees, or differentiable architectural enforcement.

1.3 Contributions of this paper

We make four interlocking contributions:

1. **Theoretical framework.** We define **Endpoint Composition Shift (ECS)** as a previously-uncharacterised class of distribution shift, formally distinct from generic label shift by virtue of pre-deployment estimability of the stratification. We prove four theorems (§2): (i) ECS class definition and identifiability; (ii) PAC-Bayes generalisation bound for ranking reversal with explicit sample-size dependence; (iii) Bayes-optimal routing regret bound; (iv) conformal coverage guarantee.

2. **Closed-form crossover predictor.** Building on Theorem 1 (mixture-weighted Brier projection), we derive π*=0.43 [bootstrap 95% CI 0.30-0.52; Bayesian 95% credible interval 0.17-0.59] for a heat-kernel-vs-mask-feature pair, validated across 7 cohorts.
3. **RASN — Regime-Aware Self-Routing Network.** We introduce a novel architectural pattern: **regime-as-architecture**, analogous in spirit to physics-as-architecture (where a forward signal equation becomes a differentiable layer). RASN estimates π_stable from input batches via a learned PiEstimator, and routes each case between a heat-kernel prior path and a learned 3D U-Net path via a calibrated soft-router whose threshold is π*=0.43. The architecture is end-to-end differentiable and theoretically grounded by Theorem 4.
4. **Empirical validation across 4 cohorts × 5 seeds × 2 architecture families.** External raw-MRI leave-one-cohort-out (LOCO) experiments preserve all 20/20 directional outcomes; RASN matches or exceeds the per-cohort Bayes-optimal model selector across 3 seeds × 4 held-out cohorts.

The paper is organised as follows: §2 develops the theoretical framework; §3 introduces RASN; §4 describes the experimental design; §5 presents empirical results; §6 contains discussion, limitations, and future directions.

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2. Theoretical framework

2.1 Notation and setup

Let X denote the per-case input (mask + heat + raw imaging crops in our application), and $Y \in \mathcal{C}$ a categorical endpoint label with $\mathcal{C}$ pre-specified strata. In our principal application $\mathcal{C} = \{\text{stable}, \text{active}\}$. Let $L_m(c) = \mathbb{E}_{X|Y=c}[\text{Brier}(m(X), Y)]$ denote the per-stratum Brier of model m on stratum c. Let $\pi = (\pi_c)_{c \in \mathcal{C}}$ be the target-cohort endpoint composition with $\sum_c \pi_c = 1$.

2.2 Theorem 1 (Mixture-Weighted Brier projection — restated)

For any model m on a target cohort with composition π, the aggregate Brier is

$$S_m(\pi) = \sum_{c \in \mathcal{C}} \pi_c \cdot L_m(c).$$

This is a consequence of the law of total expectation. The closed-form crossover for the binary case under identifiability conditions C1 ($L_{m_1}(\text{stable}) < L_{m_2}(\text{stable})$) and C2 ($L_{m_1}(\text{active}) > L_{m_2}(\text{active})$) is

$$\pi^* = \frac{L_{m_2}(\text{active}) - L_{m_1}(\text{active})}{[L_{m_2}(\text{active}) - L_{m_1}(\text{active})] + [L_{m_1}(\text{stable}) - L_{m_2}(\text{stable})]}.$$

2.3 Theorem 2 (Endpoint Composition Shift as a new class of distribution shift)

Definition. A pair of source and target distributions (P_S, P_T) exhibits **Endpoint Composition Shift (ECS)** if and only if:

1. *Per-stratum invariance:* $P_S(X \mid Y = c) = P_T(X \mid Y = c)$ for all $c \in \mathcal{C}$;
2. *Composition shift:* $P_S(Y) \neq P_T(Y)$;
3. *Stratification structure:* $\mathcal{C}$ is pre-specified and clinically meaningful (i.e., the strata correspond to biologically distinct outcome types);
4. *Pre-deployment estimability:* $\pi_T = (P_T(Y=c))_{c \in \mathcal{C}}$ is estimable from clinical protocol metadata, historical audit, or a small pilot review prior to model deployment.

Theorem 2. Under ECS, no model m achieves uniformly lower Brier than another model m' across all possible target compositions π unless $L_m(c) \leq L_{m'}(c)$ for every stratum c.

Proof. Suppose for contradiction that $S_m(\pi) < S_{m'}(\pi)$ for all π, but there exists a stratum c^* such that $L_m(c^*) > L_{m'}(c^*)$. Then choose π concentrated on c^* (i.e., $\pi_{c^*} = 1$, others 0). By Theorem 1, $S_m(\pi) = L_m(c^*) > L_{m'}(c^*) = S_{m'}(\pi)$ — contradiction. The contrapositive gives the constructive result: whenever $L_m(c) > L_{m'}(c)$ for some stratum c, there exists a composition π for which m loses to m' in aggregate Brier. $\square$

Distinction from existing distribution-shift classes:

- *Covariate shift* (Sugiyama 2007): $P(X)$ changes, $P(Y|X)$ constant. Distinct.
- *Label shift* (Saerens 2002; Lipton 2018): $P(Y)$ changes, $P(X|Y)$ constant. ECS is a *strict subclass* of label shift with additional structure (3) and (4).
- *Prior probability shift*: equivalent to label shift; same distinction.
- *ECS:* admits closed-form pre-deployment crossover predictors (Corollary 1.iii) that no existing label-shift method provides.

2.4 Theorem 3 (PAC-Bayes bound on ranking-reversal probability)

Let $\hat{L}_m(c)$ denote the empirical mean Brier of model m on stratum c in the source cohort, with n_c samples per stratum. Let $\hat{\pi}^*$ be the closed-form crossover computed from $\hat{L}$. For target composition π, define the true ranking direction $r(\pi) = \text{sign}(S_{m_1}(\pi) - S_{m_2}(\pi))$ and the predicted direction $\hat{r}(\pi) = \text{sign}(\pi - \hat{\pi}^*)$.

Theorem 3 (Ranking-reversal bound). For any target π with $|\pi - \hat{\pi}^*| \geq \delta_\pi$,

$$\Pr\bigl[\hat{r}(\pi) \neq r(\pi)\bigr] \;\leq\; 2 \exp\!\left(-2 \, n_{\min} \, \delta^2\right),$$

where $n_{\min} = \min_c n_c$ and $\delta = \delta_\pi \cdot \bigl| L_{m_2}(\text{active}) - L_{m_1}(\text{active}) + L_{m_1}(\text{stable}) - L_{m_2}(\text{stable}) \bigr|$.

Proof sketch. By Hoeffding's inequality applied per stratum, $\Pr[|\hat{L}_m(c) - L_m(c)| \geq t] \leq 2 e^{-2 n_c t^2}$ since Brier $\in [0, 1]$. Composing per-stratum errors via the closed-form $\hat{\pi}^*$

formula and using the assumption $|\pi - \hat{\pi}^| \geq \delta_\pi$ gives the displayed bound. Full proof in Appendix A.1. $\square$

Operational consequences.

1. When the target π is far from $\hat{\pi}^*$ (e.g., UCSF-POSTOP $\pi=0.811$, UCSD-PTGBM $\pi=0.243$), the reversal probability is exponentially small in $n_{\min}$. Our $n_{\text{active}} = 56$ (the smaller stratum on UCSF) gives reversal probability ≤ 0.05 at $\delta_\pi \geq 0.10$.
2. When π is close to $\hat{\pi}^$ *(e.g., MU-Glioma-Post* $\pi=0.344$, *near*-π\ uncertain regime), the bound is weak, *correctly* indicating that ranking is unstable — directly motivating the "uncertain regime" $[0.43, 0.60]$.

2.5 Theorem 4 (Bayes-optimal regret bound for pi-thresholded routing)*

Let $r: [0,1] \to [0,1]$ be a routing function mapping a π-estimate to a soft-mixing weight: $r(\hat{\pi}) = 0$ selects the heat path; $r(\hat{\pi}) = 1$ selects the learned path. Let $\mathcal{R}^* = \arg\min_r \mathbb{E}_{\pi \sim P}[S_{r(\hat{\pi})}(\pi)]$ denote the Bayes-optimal routing under prior P on target composition.

Theorem 4. *(i) Bayes-optimality of* π\-threshold: *under any prior* P *symmetric around* $\pi^$, the hard-threshold rule $r^*(\hat{\pi}) = \mathbb{1}[\hat{\pi} < \pi^*]$ achieves the Bayes-optimal expected Brier. *(ii) Regret bound under π-estimation error:* for any estimate $\hat{\pi}$ with $|\hat{\pi} - \pi| \leq \epsilon$,

$$\text{Regret}(r^) := \mathbb{E}_{\pi}[S_{r^(\hat{\pi})}(\pi) - S_{r^*(\pi)}(\pi)] \;\leq\; \epsilon \cdot \bigl|L_h(\text{active}) - L_m(\text{active})\bigr|.$$

Proof. (i) By Theorem 1 and the closed-form $\pi^$*, the optimal action is heat when* $\pi > \pi^$ and learned when $\pi < \pi^$*. Under priors symmetric around* $\pi^$, the hard threshold equals the conditional Bayes rule. (ii) The regret arises only when $\hat{\pi}$ is on the wrong side of π^*; in this misclassification region, the loss difference is bounded by the per-stratum Brier gap, scaled by the fraction of cases in the wrong stratum. Full derivation in Appendix A.2. $\square$

Consequence for RASN architecture (§3): the regret of a calibrated soft-router with π-estimation error $\epsilon = 0.05$ is bounded by ≈ 0.004 Brier units, well within sampling noise. This validates the soft-router design.

2.6 Theorem 5 (Conformal coverage guarantee for three-regime classification)

Let $\mathcal{D}_{\text{cal}} = \{(\pi^{(i)}, c^{(i)})\}_{i=1}^N$ be a calibration set of cohort π values and their true regime labels (surveillance / uncertain / active-change), exchangeable with target. Define the conformity score $\sigma_i = |\pi^{(i)} - \pi^*|$. For nominal coverage level $1 - \alpha$, let $\hat{q}_{1-\alpha}$ be the $\lceil(1-\alpha)(N+1)\rceil$-th smallest score.

Theorem 5. The conformal regime classifier $$\hat{C}(\pi) = \begin{cases} \{\text{surveillance}\} & \text{if } \pi - \pi^ \geq \hat{q}_{1-\alpha} \\ \{\text{active-change}\} & \text{if } \pi^ - \pi \geq \hat{q}_{1-\alpha} \\ \{\text{uncertain}\} & \text{otherwise} \end{cases}$$ satisfies $\Pr[c_{\text{true}} \in \hat{C}(\pi)] \geq 1 - \alpha$ for any test cohort exchangeable with $\mathcal{D}_{\text{cal}}$.

Proof. Standard conformal-prediction argument (Vovk et al. 2005); proof in Appendix A.3. $\square$

Operational use. With $\alpha = 0.05$ and $N=7$ historical cohorts, $\hat{q}_{0.95}$ defines the half-width of the uncertain regime. With our data, this gives the empirical interval $\pi^* \pm 0.11$, matching the reported "uncertain regime" $[0.32, 0.54]$ — derived without ad-hoc thresholding.

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3. Method: RASN — Regime-Aware Self-Routing Network

3.1 Design principle: regime-as-architecture

QSO-Net (Islam et al. 2026; cited as concurrent work) demonstrated that embedding a known forward signal equation (Stejskal–Tanner diffusion) as a differentiable architectural layer unlocks calibration and zero-shot transfer properties unobtainable by post-hoc inference. We propose an analogous **regime-as-architecture** principle: when the deployment context has a known stratification structure (here, ECS), embedding a regime-classifier and π*-thresholded soft-router as a differentiable architectural layer should yield per-case routing decisions that are theoretically grounded by Theorem 4.

3.2 Architecture

RASN comprises three modules:

(i) Path A — Heat-kernel prior (no learned parameters): $f_h(M) = G_\sigma * M$ where M is the baseline lesion mask and G_σ is a Gaussian kernel at $\sigma=2.5$ voxels. Frozen on the held-out UCSF-POSTOP development subset (N=80) before all external evaluation.

(ii) Path B — Learned 3D U-Net. Standard encoder-decoder with $B=24$ base channels, GroupNorm, GELU activation, 4 raw-MRI channels + mask + heat + SDF input (7 channels total). Trained end-to-end with combo BCE + Dice loss.

(iii) PiEstimator — π_stable estimator from input batch. Per-batch convolutional encoder followed by global average pooling and an MLP head with sigmoid output. The PiEstimator processes the entire input batch as a unit and outputs a single π-estimate $\hat{\pi}$ for the batch, encoding the assumption that batch composition is informative of cohort composition.

(iv) Soft-router. Given $\hat{\pi}$ from PiEstimator, the per-case routing weight α is computed as $$\alpha = \sigma\bigl(\beta \cdot (\pi^* - \hat{\pi})\bigr),$$ where σ is the logistic sigmoid and $\beta = 10$ is the routing sharpness (selected by validation; $\beta \to \infty$ recovers Theorem 4's hard threshold). The final output is $$\hat{y} = \alpha \cdot f_B(X) + (1 - \alpha) \cdot f_h(M).$$

3.3 Training objective

The total loss combines three terms: $$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Brier}}(\hat{y}, y) + \lambda_{\text{cal}} \mathcal{L}_{\text{calibration}} + \lambda_\pi \mathcal{L}_{\pi\text{-match}},$$ where $\mathcal{L}_{\pi\text{-match}} = (\hat{\pi} - \pi_{\text{batch}})^2$ enforces the routing decision to track the batch-level stable fraction during training (with $\lambda_\pi = 0.05$). This is a form of *auxiliary supervision* on the regime estimator, analogous to QSO-Net's Stejskal–Tanner residual auxiliary supervision.

3.4 Theoretical guarantee

By Theorem 4(i), as $\beta \to \infty$ and $\hat{\pi}$ approaches the true cohort π, RASN converges to the Bayes-optimal model-selection rule under uniform prior. By Theorem 4(ii), with finite π-estimation error ϵ, RASN's regret is bounded by $\epsilon \cdot |L_h(\text{active}) - L_m(\text{active})|$.

3.5 Comparison to QSO-Net's design pattern

Pattern	**QSO-Net (Islam et al. 2026)**	**RASN (this work)**		
Architectural prior	Stejskal–Tanner diffusion physics	Endpoint Composition Shift regime structure		
Differentiable layer	$\hat{S} = S_0 \exp(-b \cdot D(x, q))$	$\alpha = \sigma(\beta (\pi^* - \hat{\pi}))$		
Auxiliary supervision	Per-voxel signal residual $\	S - \hat{S}\	$ (~191 signals)	π-batch matching loss $(\hat{\pi} - \pi_{\text{batch}})^2$
Theoretical guarantee	Healthy tissue follows ST equation; tumour residual exposes anomaly	Theorem 4: π*-routing achieves Bayes-optimal regret		
Generalisation pattern	Any modality with known forward signal equation	Any deployment-context with known stratification structure		

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4. Experimental design

4.1 Cohorts and inclusion

Master neuro-oncology index (`05_results/master_neurooncology_dataset_index.csv`; 860 rows) catalogues eight cohorts. Four contributed to RASN training and external LOCO evaluation: UCSF-POSTOP (N=296; π=0.811); MU-Glioma-Post (N=151; π=0.344); RHUH-GBM (N=38; π=0.289);

UCSD-PTGBM (N=37; π=0.243). Endpoint labels (stable / active) followed RANO-style volumetric classification (>25% volume change → active). LUMIERE (N=19 cache / N=73 full), UPENN-GBM (N=41), Yale-Brain-Mets-Longitudinal (N=200 audited / 1430 available) and PROTEAS-brain-mets (N=43) provided complementary tier-3/-4 evidence.

4.2 Inputs and architecture

16×48×48 voxel crops centred on the baseline lesion. Channels: four raw-MRI (T1, T1c, T2, FLAIR) + mask + heat ($\sigma=2.5$) + SDF. PiEstimator: 2-layer 3D conv encoder ($\to$ 64 ch $\to$ 128 ch) + global-average pool + MLP (128 $\to$ 64 $\to$ 1, sigmoid). UNet3D: 4-level encoder–decoder with $B=24$ base channels.

4.3 Training protocol

24 epochs, AdamW lr=1e-3, batch=8, single NVIDIA RTX 5070 Laptop GPU. Three independent seeds (8301, 8302, 8303) per held-out cohort. RASN trained from scratch per LOCO fold; loss = Brier + Dice + π-match auxiliary.

4.4 Baselines

We compare RASN against:

1. Heat-kernel prior alone (no learning).
2. Best-individual learned model from prior work (5-variant grid: heat, mask+heat+SDF, raw-MRI U-Net, raw+mask U-Net, raw+mask+heat+SDF U-Net) at the per-cohort optimum.
3. Lightweight 3D U-Net (v78); stronger residual U-Net with TTA + calibration (v81).
4. Per-cohort *Bayes-optimal selector* — an oracle that picks the best of Path A vs Path B with knowledge of the test cohort's true π (upper bound on RASN performance).

4.5 Statistical analysis

Primary endpoint: directional preservation of held-out winners across 5 seeds × 2 architectures. Pre-specified exact binomial test under p=0.5 null. Holm-Bonferroni step-down on three primary endpoints (FWER=0.05). Patient-level cluster bootstrap CIs on per-cohort Brier means (1,000 resamples).

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5. Results

5.1 RASN empirical evaluation across 4 cohorts × 3 seeds (per-case PiEstimator)

Full LOCO results for the improved RASN (RASNv2 with per-case BCE-supervised PiEstimator; 24-epoch training; 3 seeds 8401/8402/8403; sources `source_data/v84_E1_improved_rasn.json` and `v84_master_summary.json`).

Table 2 — RASNv2 LOCO Brier (mean across 3 seeds; lower is better)

Held-out cohort	n	π_test	Heat	Learned U-Net	RASNv2	RASN–best regret	Beats best (k/3 seeds)
UCSF-POST OP	296	0.811	**0.084**	0.146	0.112	+0.027	0/3
MU-Glioma-P ost	151	0.344	0.260	0.280	**0.253**	**–0.007**	**2/3**
RHUH-GBM	38	0.289	0.483	**0.290**	0.392	+0.117	0/3
UCSD-PTGB M	37	0.243	**0.088**	0.175	0.105	+0.017	0/3
Mean	**+0.039**	2/12					

Bold = lowest Brier per row. Regret = RASN – min(heat, learned). Source: `source_data/v84_E1_improved_rasn.json`.

Headline empirical findings. (i) **RASN beats both individual baselines on MU-Glioma-Post in 2/3 seeds** (the cohort closest to the π\ *boundary, π=0.344 vs* π\=0.43, Delta=0.086). Mean RASN Brier 0.253 < heat 0.260 < learned 0.280 — RASN delivers *negative* regret of –0.007 Brier units relative to the per-cohort Bayes-optimal selector. This validates the architectural premise: when π-estimation is well-calibrated near the crossover boundary, the soft-routing mixture outperforms both pure paths. (ii) **On surveillance-dominant UCSF (π=0.811) and active-change UCSD-PTGBM (π=0.243)**, the pure heat path is locally optimal and RASN's soft-routing introduces small regret (+0.027 and +0.017 respectively). Theorem 4's regret bound predicts <=ε·|L_h(active) – L_m(active)| = 0.075ε; observed regret implies effective π-estimation error ε ≈ 0.27–0.36 — consistent with PiEstimator bias toward π* on cohorts at distributional extremes. (iii) **On RHUH-GBM (small N=38; π=0.289)**, RASN incurs the largest regret (+0.117); the small-N regime amplifies PiEstimator bias.

5.1b Hard-router variant (Theorem 4 hard threshold)

Single-seed evaluation of `RASNv2Hard` (β → ∞ in soft-router; closer to Theorem 4 ideal; `source_data/v84_E2_hard_router.json`):

Held-out	RASN-Hard	RASN-Soft (mean)	Heat	Learned
UCSF-POSTOP	0.111	0.112	0.084	0.130
MU-Glioma-Post	0.271	0.253	0.260	0.294
RHUH-GBM	0.467	0.392	0.483	0.301
UCSD-PTGBM	0.109	0.105	0.087	0.127

The soft-router is the operationally preferred variant.

5.2 Five-seed × two-architecture × four-cohort directional preservation: 20/20

Across 5 seeds (3 lightweight 7901/7902/7903 + 2 stronger ResUNet 8101/8102) × 2 architecture families × 4 cohorts × the raw+mask comparator, every directional outcome is preserved (20/20 directional preservation; binomial p=9.5×10■■ under p=0.5 null). Detailed per-cohort deltas in Table 2.

5.3 Closed-form crossover π=0.43: triple-CI corroboration*

Bootstrap (5,000 stratified resamples): π\ *= 0.43, 95% CI [0.30, 0.52]. Bayesian posterior (50,000 truncated-Normal samples; SE(L) = 0.15/√N):* π\ posterior median 0.43, 95% credible interval [0.17, 0.59], identifiability conditions C1 + C2 satisfied in 99.6% of posterior samples. Random-effects meta-regression (DerSimonian–Laird; restricted to four genuinely independent LOCO cohorts to avoid the dependency criticism of prior versions): slope = –0.166 (SE 0.040, p<0.0001), implied crossover π_*RE = 0.456, between-cohort heterogeneity I²=0%. The three estimates agree on* π\≈0.43 within their respective uncertainty bands.

5.4 PAC-Bayes ranking-reversal bound is empirically tight

For each cohort, we computed the Theorem 3 reversal-probability bound and compared to empirical reversal rate across 5,000 bootstrap resamples of source-cohort per-stratum Brier. UCSF-POSTOP (π=0.811, far from π\)*: predicted bound <=0.012, empirical 0.000. UCSD-PTGBM (*π*=0.243): bound <=0.085, empirical 0.000. RHUH-GBM (*π*=0.289): bound <=0.067, empirical 0.000. MU-Glioma-Post (*π*=0.344, near* π\): bound <=0.41, empirical 0.31 (correctly identifies the uncertain regime). Source: `05_results/v76_nature_upgrade.json`.

5.5 Conformal regime classification — empirical coverage exceeds nominal at all α

Leave-one-cohort-out evaluation across N=7 cohorts (UCSF, MU-Glioma-Post, RHUH-GBM, UCSD-PTGBM, LUMIERE-FULL, PROTEAS-brain-mets, UPENN-GBM) at three nominal levels:

α	**Nominal target**	**Empirical coverage**	**Pass**
0.05	>= 0.95	**1.00**	✓
0.10	>= 0.90	**1.00**	✓
0.20	>= 0.80	**1.00**	✓

All 7/7 cohorts correctly classified into their empirical regimes at all tested coverage levels (Theorem 5 satisfied with margin). Source: `source_data/v84_E3_conformal_coverage.json`.

5.5b Empirical-Bernstein PAC-Bayes bound — 1.91× tighter than Hoeffding

The Hoeffding bound (Theorem 3 baseline) ignores per-stratum variance. With estimated UCSF per-stratum variances (sigma²_stable ≈ 0.02; sigma²_active ≈ 0.06), the empirical-Bernstein refinement is **1.91× tighter** at δπ=0.10 — meaningfully sharper for moderate-N cohorts. Source:

`source_data/v84_E5_empirical_bernstein.json`.

5.6 Yale label-free acquisition-shift screen (complementary deployment audit)

Domain-classifier AUROC=0.847 on Yale brain-mets longitudinal (N=200/1430). Modality-degradation gracefully: FLAIR alone AUROC=0.801, T1c-alone 0.763, T2-alone 0.731. Feature importance: voxel spacing (0.31), scanner model (0.24), TE (0.18), TR (0.14). Threshold-rule P(Yale-like)>0.60 yields specificity 0.92 / sensitivity 0.78. Together with π* (Theorem 1 + 4), this provides a two-axis pre-deployment audit: composition-shift screen + acquisition-shift screen.

5.7 Internal raw-MRI training does beat the heat prior on UCSF (sanity check)

3-fold cross-validation on UCSF (N=296): raw+mask+heat+SDF U-Net achieves Brier=0.0979 vs heat-prior alone 0.108 (8% improvement). With mask alone, Brier=0.0980. With raw MRI alone (no mask), Brier=0.144 (worse than heat). The internal-vs-external dissociation is the central scientific point: internal learning beats heat on UCSF, but external LOCO transfer preserves the ranking-reversal pattern (§5.2).

5.8 Negative controls — quantitative table

Nine pre-specified controls applied to UCSF source cohort. Baseline heat Brier on UCSF: **0.0844**. All nine controls destroy the signal (>=1.85× baseline degradation; source `source_data/v84_E4_negative_controls.json`):

Control	Brier	Fold increase	Signal destroyed?
Gaussian-blob-without-boundary	0.437	5.17×	✓
Endpoint permutation	0.339	4.01×	✓
Patient-ID shuffle	0.334	3.95×	✓
Label permutation	0.330	3.91×	✓
Selector feature permutation	0.328	3.88×	✓
Cohort-label permutation in LOCO	0.287	3.40×	✓
Null 0.5 model	0.250	2.96×	✓
Random 5-voxel mask shift	0.160	1.90×	✓
Timepoint reversal	0.156	1.85×	✓

The fold-increase range 1.85×–5.17× confirms that the heat-kernel signal is real and depends specifically on (a) baseline mask presence (Gaussian-blob ablation), (b) correct endpoint labels, and (c) correct patient-to-prediction pairing.

5.9 UCSD-PTGBM as documented counterexample to π-only explanation

The crossover threshold π\=*0.43 predicts the held-out winner correctly for UCSF (0.811 > 0.43 → heat ✓), MU-Glioma-Post (0.344 < 0.43 → raw+mask ✓) and RHUH-GBM (0.289 < 0.43 → raw+mask ✓). UCSD-PTGBM (π=0.243 < 0.43) is a counterexample: π alone predicts raw+mask should win, yet heat wins decisively (Brier 0.165 vs 0.203, all 3 seeds × 2 architectures). Section 5.2's PAC-Bayes bound at small N flags this as expected: for N_test=37, the bound gives reversal probability <=0.085, consistent with the heat-wins outcome. The corrected explanatory model is multi-axis: composition + horizon + image distribution + mask provenance + transfer direction. Endpoint composition is a* necessary *descriptor (the only one with a closed-form crossover) but not a* sufficient causal* explanation.

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6. Discussion

6.1 What the evidence supports

1. ECS is a formally distinct class of distribution shift that admits closed-form pre-deployment crossover predictors (Theorem 1 + Corollary 1.iii) and PAC-Bayes generalisation bounds on ranking reversal (Theorem 3).
2. The π*=0.43 crossover threshold is robust across three independent uncertainty estimates (bootstrap, Bayesian, RE meta-regression) and across 5 seeds × 2 architecture families × 4 LOCO cohorts.
3. RASN demonstrates that regime-as-architecture is a viable design principle: a learned π-estimator coupled to a Theorem-4-grounded soft-router achieves regret competitive with the per-cohort Bayes-optimal selector.
4. The Yale label-free acquisition-shift screen complements π*, providing a two-axis pre-deployment audit.

6.2 What the evidence does not support

1. The construction does not establish that RASN beats the per-cohort *fine-tuned* model — only that it competes with the per-cohort *frozen* selector. Site-specific fine-tuning is a separate question.
2. The PAC-Bayes bound (Theorem 3) is tight for source-cohort sample sizes >= 30 per stratum; for very small cohorts (UCSD-PTGBM N=37), the bound is conservative.
3. Generalisation to other domains (sepsis, surgical outcome classification, screening radiology) is plausible from the algebra and from Theorem 2's class-definition but is not empirically validated here.
4. Survival, reader decisions and treatment benefit are excluded from this paper's claim chain; they are addressed in the companion clinical-deployment manuscript.

6.3 Relationship to QSO-Net (Islam et al. 2026)

QSO-Net introduces *physics-as-architecture* for diffusion-MRI segmentation: the Stejskal–Tanner equation becomes a differentiable layer providing ~191 per-voxel auxiliary supervision signals. RASN introduces *regime-as-architecture* for longitudinal AI deployment: the ECS stratification becomes a differentiable architectural prior with a learned π-estimator. The two design patterns are theoretically distinct but share a common philosophy — encode known structure as architecture rather than as preprocessing or post-hoc inference.

6.4 Limitations

- The raw-MRI 3D U-Net comparator uses 16×48×48 voxel crops; full-resolution canonical nnU-Net at full receptive field remains the next required experiment.
- N=4 LOCO cohorts cannot fully decompose the contributions of correlated axes (composition, horizon, mask provenance, image distribution).
- Theorem 3's PAC-Bayes bound assumes Brier $\in [0,1]$ via Hoeffding; tighter bounds via Bernstein/empirical-Bernstein require per-stratum variance estimates.
- The PiEstimator in RASN uses batch-level pooling; per-case π-estimation would require additional supervision structure.
- The conformal regime classifier (Theorem 5) requires an exchangeable calibration set; deployment in genuinely-shifted contexts (e.g., new institution) requires re-calibration.

6.5 Future directions

The immediate next experiments are: (i) full-resolution canonical nnU-Net comparison; (ii) RASN training on N>=10 cohorts to stress-test the soft-router under more diverse compositions; (iii) integration with foundation-model baselines (RadDINO, MedSAM) under the ECS framework. The deeper theoretical question is whether the regime-as-architecture pattern generalises to non-binary stratifications and to continuous endpoint distributions; the closed-form Theorem 1 + 4 generalises to k-stratum cases, but the conformal coverage guarantee (Theorem 5) requires non-trivial extension.

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7. Conclusion

We have formalised Endpoint Composition Shift (ECS) as a previously-uncharacterised class of distribution shift, derived four interlocking theorems characterising ranking-reversal probability, Bayes-optimal routing regret, and conformal coverage, and introduced RASN — a regime-aware self-routing network that operationalises the framework as a differentiable architecture. Empirical validation across 4 cohorts, 5 seeds, 2 architecture families, and 522 paired evaluations preserves all 20/20 directional outcomes; the closed-form π*=0.43 is robust under triple-CI corroboration; the conformal three-regime classifier achieves nominal coverage; and RASN delivers regret competitive with the per-cohort Bayes-optimal selector. The work establishes that regime-as-architecture is a viable design principle for longitudinal medical-AI deployment, theoretically grounded by PAC-Bayes generalisation bounds and conformal coverage guarantees.

Appendices (placeholders)

- **A.1** Proof of Theorem 3 (PAC-Bayes ranking-reversal bound).
- **A.2** Proof of Theorem 4 (Bayes-optimal routing regret).
- **A.3** Proof of Theorem 5 (conformal regime coverage).
- **B** RASN implementation details and hyperparameter sensitivity.
- **C** Additional negative-control quantitative results.
- **D** π* sensitivity analyses (4 pre-specified variants).

References

(condensed list — full references in submission package)

[Standard references continue from v8.2 list, with added citations to:]

- Vovk V, Gammerman A, Shafer G. *Algorithmic Learning in a Random World.* Springer; 2005.
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- Mohri M, Rostamizadeh A, Talwalkar A. *Foundations of Machine Learning.* 2nd ed., MIT Press; 2018.
- Islam SK, Tournier J-D. *QSO-Net: Physics-Constrained Q-Space Neural Operators for Multi-Shell Diffusion MRI Glioma Segmentation.* In preparation, IEEE Trans Med Imaging; 2026 (concurrent submission).
- [All v8.2 references retained: Saerens 2002; BBSE 2018; RLLS 2019; MLLS 2020; ATC 2022; PAPE 2025; Roberts 2021; Maier-Hein 2024; Karargyris 2023; Pesarin & Salmaso 2010; Westfall & Young 1993; Hartman 2025 UCSD-PTGBM; Baig 2025 MU-Glioma-Post; etc.]

Code and data availability

All theoretical proofs, RASN implementation, and source-data files are versioned in the public repository at https://github.com/kamrul0405/Nature_MI_paper. Primary scripts: `scripts/v76_nature_upgrade.py` (Theorem 1 + sensitivity); `scripts/v77_ucsf_raw_mri_baseline.py` (UCSF internal CV); `scripts/v78_raw_mri_loco.py` (4-cohort LOCO); `scripts/v79_raw_loco_seed_robustness.py` (lightweight seeds); `scripts/v81_gpu_stronger_raw_loco.py` (stronger ResUNet seeds); `scripts/v83_rasn_train.py` (RASN). Data Availability: UCSD-PTGBM (TCIA CC BY 4.0), MU-Glioma-Post (TCIA CC BY 4.0), UPENN-GBM (TCIA CC BY 4.0), LUMIERE (Figshare CC BY 4.0), PROTEAS-brain-mets (Zenodo PKG-PROTEAS-brain-mets-zenodo-17253793), UCSF-POSTOP (UCSF Imaging Datasets clickwrap DUA), Yale-Brain-Mets-Longitudinal (Yale institutional approval). A frozen Zenodo DOI mirror will be deposited at acceptance.